Qubits, Measurement, Postulates of Quantum Mechanics and Math Examples

Samya Muhuri

For different books, symbolic representations are different. For each case, book and page no is given
(Except math examples)

Different Physical Types of Qubits

Photonic Qubits (Light / Photons)

Pros:

Very low decoherence

Ideal for communication

Room temperature operation

Cons:

Hard to make photons interact (two-qubit gates difficult)

Superconducting Qubits

Pros:

Fast gate speeds

Scalable on chips

Leading platform today

Cons:

Requires millikelvin temperatures

Different Physical Types of Qubits

Trapped Ion Qubits

Pros:

Extremely high fidelity

Long coherence times

Cons:

Slower gates

Harder to scale physically

Spin Qubits (Electrons or Nuclei)

Pros:

Compatible with semiconductor tech

Potentially very scalable

Cons:

Control is challenging

Different Physical Types of Qubits

Neutral Atom Qubits

Pros:

Large arrays possible

Good coherence

Cons:

Complex laser control

Quantum Probability

Quantum probability theory is the mathematical framework used to describe probabilities in quantum mechanics. The axioms of quantum probability are based on the principles of quantum mechanics and provide rules for assigning probabilities to measurement outcomes. Here are the key axioms of quantum probability:

State Axiom: The state of a quantum system is described by a state vector, often denoted as $|\psi\rangle$, which belongs to a complex vector space known as the Hilbert space.

Quantum Probability

Probability Axiom: The probability of obtaining a particular measurement outcome is given by the square of the absolute value of the probability amplitude associated with that outcome. For a state $|\psi\rangle$ and a measurement corresponding to a Hermitian operator M, the probability of obtaining outcome m is given by:

$$P(m) = |\langle m|\psi\rangle|^2$$

Here, $|m\rangle$ is an eigenstate of the operator M associated with eigenvalue m, and $\langle m|\psi\rangle$ is the inner product of $|m\rangle$ and $|\psi\rangle$.

Quantum Probability

Normalization Axiom: The state vector $|\psi\rangle$ must be normalized, meaning that the sum of the probabilities for all possible outcomes must equal 1:

$$\sum_m P(m) = 1$$

This reflects the idea that the system must be in one of its possible states after a measurement.

Quantum Probability

Projection Postulate: After a measurement is performed, the state of the system "collapses" into the eigenstate corresponding to the observed outcome.

Mathematically, if the measurement outcome is m, the state $|\psi\rangle$ collapses to the normalized projection of $|\psi\rangle$ onto the subspace spanned by the eigenstate $|m\rangle$.

These axioms are fundamental to the mathematical formulation of quantum mechanics and provide a probabilistic description of physical systems at the quantum level. They ensure consistency with experimental observations and predictions.

State collapse

- In quantum mechanics, particles are described by wave functions that represent a range of possible states.
- When a measurement is made on a quantum system, the act of measurement causes the system to "collapse" into one of its possible states.
- Before measurement, a particle exists in a superposition of multiple states, meaning it can be in several states simultaneously.
- However, the act of measurement forces the particle to assume a definite state.
- The collapse of the wave function represents the transition from this superposition of possibilities to a specific, observable state.

Pure State

1. A pure state is described by a ket vector $|\psi\rangle$ in a Hilbert space.
2. It represents a system that is in a perfectly definite and well-defined state.
3. The probability distribution associated with a measurement on a pure state is sharply peaked, meaning that a measurement will yield a specific outcome with certainty.
4. The density operator for a pure state is given by $\rho = |\psi\rangle\langle\psi|$.

Mixed State

1. A mixed state, on the other hand, is a statistical ensemble of pure states. It arises when a system is in a state of partial knowledge or uncertainty.

2. The state is described by a density operator ρ, which is a Hermitian, positive semi-definite matrix with a trace equal to 1.

3. The probability distribution associated with a measurement on a mixed state is a combination of the probabilities associated with the pure states in the ensemble.

4. Mathematically, a mixed state can be represented as a statistical mixture of pure states: $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$, where p_i is the probability of the system being in the pure state $|\psi_i\rangle$.

5. Unlike a pure state, a mixed state reflects an element of uncertainty or lack of knowledge about the exact quantum state of the system.

Density operators

The density operator is denoted by the symbol ρ, and for a pure state $|\psi\rangle$, it is given by:

$$\rho = |\psi\rangle\langle\psi|$$

Here, $|\psi\rangle$ is the state vector (ket) of the system, and $\langle\psi|$ is its conjugate transpose, also known as the bra vector.

For a mixed state, which is a statistical ensemble of pure states with associated probabilities, the density operator is a weighted sum of the individual density operators corresponding to each pure state:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

Here, p_i is the probability of the system being in the pure state $|\psi_i\rangle$.

Properties of Density Operators

Hermitian: The density operator is always Hermitian, meaning it is equal to its own conjugate transpose: $\rho^\dagger = \rho$.

Trace: The trace of the density operator is equal to 1 for a normalized state. The trace is a measure of the total probability in the system: $Tr(\rho) = 1$.

Expectation Values: The expectation value of an observable A in the state described by the density operator ρ is given by $\langle A \rangle = Tr(\rho A)$.

Evolution: The density operator evolves in time according to the von Neumann equation: $\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho]$, where H is the Hamiltonian of the system.

The density operator is a powerful tool as it allows for the representation of both pure and mixed states within the same formalism. It is particularly useful in dealing with statistical ensembles of quantum systems and provides a convenient way to describe the partial knowledge or uncertainty about the quantum state of a system.

Projection Operators

The number of projection operators is determined by the dimension of the Hilbert space

If the dimension of the Hilbert space is d and there are m projection operators, it must be true that

$$m \leq d$$

For example, if we are talking about a qubit where $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the dimension of the space is 2, and the projection operators corresponding to the mutually exclusive measurement results $|0\rangle$ and $|1\rangle$ are

$$P_0 = |0\rangle\langle 0|, \quad P_1 = |1\rangle\langle 1|$$

Phase

A quantum state can have complex amplitudes:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

If we multiply by a complex number of unit magnitude:

$$e^{i\theta}$$

we change the **phase**.

Global Phase

Multiplying the *entire state* by the same phase:

$$\boxed{|\psi\rangle \to e^{i\theta}|\psi\rangle}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Multiply by $e^{i\pi/3}$:

$$|\psi'\rangle = \frac{e^{i\pi/3}}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$\boxed{\text{Global phase has NO observable effect}}$$

Relative Phase

Phase difference **between components** of the superposition.

$$\boxed{|\psi\rangle = \alpha|0\rangle + e^{i\phi}\beta|1\rangle}$$

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi}|1\rangle)$$

They differ only by **relative phase π**.

$$\boxed{\text{Relative phase DOES affect outcomes}}$$

Postulates of quantum mechanics (Pg:80 in Nielsen Book)

- Quantum mechanics is a mathematical framework for the development of physical theories
- Postulate 1: Associated to any isolated physical system is a complex vector space with inner product (that is, a Hilbert space) known as the state space of the system. The system is completely described by its state vector, which is a unit vector in the system's state space.
- Suppose $|0\rangle$ and $|1\rangle$ form an orthonormal basis for that state space. Then an arbitrary state vector in the state space can be written

$$|\psi\rangle = a|0\rangle + b|1\rangle,$$

- where a and b are complex numbers. The condition that $|\psi\rangle$ be a unit vector, $\langle\psi|\psi\rangle = 1$, is therefore equivalent to $|a|^2 + |b|^2 = 1$. The condition $\langle\psi|\psi\rangle = 1$ is often known as the normalization condition for state vectors.

Qudit State

- Qutrits and Qudits denote quantum system with more than two levels.
- Qutrits (3 level system)
- Qudits (general d-level systems)

The general state of a qudit can be expressed as:

$$|\psi\rangle = \sum_{j=0}^{d-1} c_j \, |j\rangle \, ,$$

where $c_j \in \mathbb{C}$. This state's normalization constraint is given by:

$$\langle\psi|\psi\rangle = \sum_{j=0}^{d-1} |c_j|^2 = 1.$$

Postulates of quantum mechanics

- Postulate 2: The evolution of a closed quantum system is described by a unitary transformation. That is, the state $|\psi\rangle$ of the system at time t_1 is related to the state $|\psi'\rangle$ of the system at time t_2 by a unitary operator U which depends only on the times t_1 and t_2,

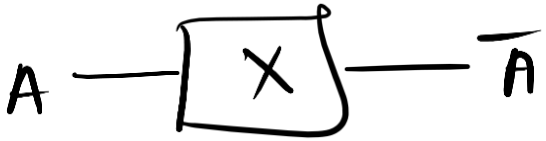
$$|\psi'\rangle = U|\psi\rangle \ .$$

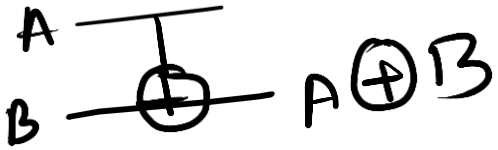
Closed quantum systems are systems where the energy of the system is conserved in time.

Gates are the examples of Unitary operators.
Boolean Representation of Quantum Gates

Classical → Quantum equivalents:

Classical	Reversible Quantum
NOT	X gate
XOR	CNOT
AND	Toffoli (CCX)
OR	Built from AND + XOR
NAND	Toffoli + X



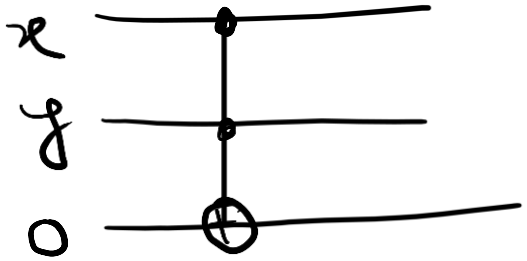


Classical function:

$$f(x, y) = x \wedge y$$

Quantum reversible version:

$$|x, y, 0\rangle \to |x, y, xy\rangle$$



Gate	Symbol	Boolean Representation
X (NOT)	$\lvert x\rangle$ — $\boxed{X}$ — $\lvert\bar{x}\rangle$	$\lvert x\rangle \mapsto \lvert\bar{x}\rangle$
CNOT	$\lvert x\rangle$ — $\bullet$ — $\lvert x\rangle$ $\lvert y\rangle$ — $\oplus$ — $\lvert x \oplus y\rangle$	$\lvert xy\rangle \mapsto \lvert x(x \oplus y)\rangle$
Z	$\lvert x\rangle$ — $\boxed{Z}$ — $(-1)^x \lvert x\rangle$	$\lvert x\rangle \mapsto (-1)^x \lvert x\rangle$
CZ	$\lvert x\rangle$ — $\bullet$ — $\lvert x\rangle$ $\lvert y\rangle$ — $\boxed{Z}$ — $(-1)^{xy} \lvert y\rangle$	$\lvert xy\rangle \mapsto (-1)^{xy} \lvert xy\rangle$
H	$\lvert x\rangle$ — $\boxed{H}$ — $\frac{1}{\sqrt{2}}(\lvert 0\rangle + (-1)^x \lvert 1\rangle)$	$\lvert x\rangle \mapsto \frac{1}{\sqrt{2}}(\lvert 0\rangle + (-1)^x \lvert 1\rangle)$
Y	$\lvert x\rangle$ — $\boxed{Y}$ — $i(-1)^x \lvert\bar{x}\rangle$	$\lvert x\rangle \mapsto i(-1)^x \lvert\bar{x}\rangle$
$P(\phi)$	$\lvert x\rangle$ — $\boxed{e^{i\phi}}$ — $e^{ix\phi} \lvert x\rangle$	$\lvert x\rangle \mapsto e^{ix\phi} \lvert x\rangle$
U	$\lvert x\rangle$ — $\boxed{U}$ — $u_{xx} \lvert x\rangle + u_{\bar{x}x} \lvert\bar{x}\rangle$	$\lvert x\rangle \mapsto u_{xx} \lvert x\rangle + u_{\bar{x}x} \lvert\bar{x}\rangle$
CU	$\lvert x\rangle$ — $\bullet$ — $\lvert x\rangle$ $\lvert y\rangle$ — $\boxed{U}$ — $U^x \lvert y\rangle$, or $u^x_{yy} \lvert y\rangle + xu_{\bar{y}y} \lvert\bar{y}\rangle$	$\lvert xy\rangle \mapsto \lvert x\rangle U^x \lvert y\rangle$ $\lvert xy\rangle \mapsto u^x_{yy} \lvert xy\rangle + xu_{\bar{y}y} \lvert x\bar{y}\rangle$
Toffoli (CCNOT)	$\lvert x\rangle$ — $\bullet$ — $\lvert x\rangle$ $\lvert y\rangle$ — $\bullet$ — $\lvert y\rangle$ $\lvert z\rangle$ — $\oplus$ — $\lvert xy \oplus z\rangle$	$\lvert xyz\rangle \mapsto \lvert xy(xy \oplus z)\rangle$
SWAP	$\lvert x\rangle$ — $\bullet$ — $\lvert y\rangle$ $\lvert y\rangle$ — $\times$ — $\lvert x\rangle$	$\lvert xy\rangle \mapsto \lvert yx\rangle$
Fredkin (CSWAP)	$\lvert x\rangle$ — $\bullet$ — $\lvert x\rangle$ $\lvert y\rangle$ — $\times$ — $\lvert \bar{x}y \oplus xz\rangle$ $\lvert z\rangle$ — $\times$ — $\lvert xy \oplus \bar{x}z\rangle$	$\lvert xyz\rangle \mapsto \lvert x(\bar{x}y \oplus xz)(xy \oplus \bar{x}z)\rangle$

Probabilities and measurements

Postulate 3: Quantum measurements are described by a collection $\{M_m\}$ of *measurement operators*.

These are operators acting on the state space of the system being measured.

The index *m* refers to the measurement outcomes that may occur in the experiment.

If the state of the quantum system is $|\psi\rangle$ immediately before the measurement then the probability that result *m* occurs is given by

$$p(m) = \langle \Psi | M_m^\dagger M_m | \Psi \rangle$$

Probabilities and measurements

the state of the system after the measurement is

$$\frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}} .$$

The measurement operators satisfy the *completeness equation*

$$\sum_m M_m^\dagger M_m = I .$$

The completeness equation expresses the fact that probabilities sum to one:

$$1 = \sum_m p(m) = \sum_m \langle\psi|M_m^\dagger M_m|\psi\rangle$$

Projective Measurement

A projective measurement is described by an observable, M, a Hermitian operator on the state space of the system being observed. The observable has a spectral decomposition,

$$M = \sum_m m P_m$$

where P_m is the projector onto the eigenspace of M with eigenvalue m. The possible outcomes of the measurement correspond to the eigenvalues, m, of the observable. Upon measuring the state $|\psi\rangle$, the probability of getting result m is

$$p(m) = \langle\psi|P_m|\psi\rangle$$

Hermitian operator

An **Hermitian operator** is the physicist's version of an object that mathematicians call a self-adjoint operator. It is a linear operator on a vector space *V* that is equipped with positive definite inner product.

$$\langle \Phi | \Psi \rangle \in \mathbb{C}, \quad \Phi, \Psi \in V.$$

Here "Φ is in the bra" and "Ψ is in the ket".

Projective Measurement

Given that outcome m occurred, the state of the quantum system immediately after the measurement is

$$\frac{P_m|\psi\rangle}{\sqrt{p(m)}}$$

Postulate

Postulate 4: The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n, and system number i is prepared in the state ρ_i, then the joint state of the total system is $\rho_1 \otimes \rho_2 \otimes \ldots \otimes \rho_n$.

All Postulates (Page no:102 in Nielsen Book)

Postulate 1: Associated to any isolated physical system is a complex vector space with inner product (that is, a Hilbert space) known as the *state space* of the system. The system is completely described by its *density operator*, which is a positive operator ρ with trace one, acting on the state space of the system. If a quantum system is in the state ρ_i with probability p_i, then the density operator for the system is $\sum_i p_i \rho_i$.

Postulate 2: The evolution of a *closed* quantum system is described by a *unitary transformation*. That is, the state ρ of the system at time t_1 is related to the state ρ' of the system at time t_2 by a unitary operator U which depends only on the times t_1 and t_2,

$$\rho' = U \rho U^\dagger. \tag{2.158}$$

All Postulates (Page no:103 in Nielsen Book)

Postulate 3: Quantum measurements are described by a collection $\{M_m\}$ of *measurement operators*. These are operators acting on the state space of the system being measured. The index m refers to the measurement outcomes that may occur in the experiment. If the state of the quantum system is ρ immediately before the measurement then the probability that result m occurs is given by

$$p(m) = \mathrm{tr}(M_m^\dagger M_m \rho), \tag{2.159}$$

and the state of the system after the measurement is

$$\frac{M_m \rho M_m^\dagger}{\mathrm{tr}(M_m^\dagger M_m \rho)}. \tag{2.160}$$

The measurement operators satisfy the *completeness equation*,

$$\sum_m M_m^\dagger M_m = I. \tag{2.161}$$

Postulate 4: The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n, and system number i is prepared in the state ρ_i, then the joint state of the total system is $\rho_1 \otimes \rho_2 \otimes \ldots \rho_n$.

Matrix Definitions

A^*: complex conjugate of A

A^T: transpose of A

$A^\dagger$: Hermitian conjugate or adjoint of A, $A^\dagger = \left(A^{\mathrm{T}}\right)^*$

Hermitian (self-adjoint) matrix: $H^\dagger = H$

Unitary matrix: $U^\dagger = U^{-1}$ or $U^\dagger U = I$

Normal matrix: $AA^\dagger = A^\dagger A$ (includes Hermitian matrix and unitary matrix)

Trace: $\mathrm{tr}\, A \equiv \sum_i a_{ii}$

Determinant: $\det A = |A|$

Commutator: $[A, B] \equiv AB - BA$

Anti-commutator: $\{A, B\} \equiv AB + BA$

Direct Sum: $A \oplus B$ (block diagonal matrix of A and B)

Maths on Measurement

- Chapter 6, Book: Quantum Computing Explained, David Mcmahon
- Page: 125 Onwards

Maths on Linear Algebra

Chapter 1 and 2, Book: Quantum Computing for Computer Scientists
Note: See only the maths
Topics: Complex numbers, Complex vector spaces, Hermitian and Unitary Matrices (Pg:62 onwards), Tensor product (pg:71 onwards), Eigenvalue and Eigenvector, Density Matrix, Trace of Matrix.

Operators as Matrices

Example operator A:

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$$

Act on state:

$$|\psi\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$A|\psi\rangle = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

Hermitian Operators

$$\boxed{A^\dagger = A}$$

(† = conjugate transpose)

Example 1

$$H = \begin{pmatrix} 2 & i \\ -i & 3 \end{pmatrix}$$

Compute conjugate transpose:

$$H^\dagger = \begin{pmatrix} 2 & i \\ -i & 3 \end{pmatrix} = H$$

Example 2

$$B = \begin{pmatrix} 0 & 2+i \\ 2-i & 5 \end{pmatrix}$$

Take conjugate transpose:

1. Conjugate:

$$\begin{pmatrix} 0 & 2-i \\ 2+i & 5 \end{pmatrix}$$

2. Transpose:

$$B^\dagger = \begin{pmatrix} 0 & 2+i \\ 2-i & 5 \end{pmatrix} = B$$

Unitary Operators (Quantum Gates)

$$\boxed{U^\dagger U = I}$$

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Check:

$$H^\dagger H = I$$

So H is unitary.

Eigenvalues and Eigenvectors

Eigenvalues λ satisfy: $\boxed{A|v\rangle = \lambda|v\rangle}$ Non-zero vector $\mathbf{v}$ = eigenvector.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Find Eigenvalues, solve

$$\det(A - \lambda I) = 0$$

Step 1:

$$A - \lambda I = \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix}$$

Step 2:

$$\det = (2-\lambda)^2 - 1$$

Expand:

$$= \lambda^2 - 4\lambda + 3$$

Set to zero:

$$\lambda^2 - 4\lambda + 3 = 0$$

Factor:

$$(\lambda - 1)(\lambda - 3) = 0$$

Step 3:

Eigenvalues

$$\boxed{\lambda_1 = 1, \quad \lambda_2 = 3}$$

Step 4: Find Eigenvector for $\lambda = 3$

Solve:

$$(A - 3I)\mathbf{v} = 0$$

$$A - 3I = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$$

Let:

$$\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Equation:

$$-x + y = 0 \Rightarrow y = x$$

Choose $x = 1$:

$$\boxed{v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}}$$

Step 5: Eigenvector for $\lambda = 1$

$$A - I = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Equation:

$$x + y = 0 \Rightarrow y = -x$$

Choose $x = 1$:

$$\boxed{v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}}$$

Step 6: Verification

For $\lambda = 3$

$$A\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}3\\3\end{pmatrix} = 3\begin{pmatrix}1\\1\end{pmatrix}$$

For $\lambda = 1$

$$A\begin{pmatrix}1\\-1\end{pmatrix} = \begin{pmatrix}1\\-1\end{pmatrix} = 1\begin{pmatrix}1\\-1\end{pmatrix}$$

Eigenvalues and Eigenvectors

$$\boxed{A|v\rangle = \lambda|v\rangle}$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Eigenvectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Eigenvalues:

$$Z|0\rangle = +1|0\rangle$$

$$Z|1\rangle = -1|1\rangle$$

Spectral Decomposition

For a Hermitian operator A with eigenvalues λ_i and orthonormal eigenvectors $|v_i\rangle$:

$$\boxed{A = \sum_i \lambda_i \, |v_i\rangle\langle v_i|}$$

λ_i → eigenvalues (real for Hermitian operators)

$|v_i\rangle$ → eigenvectors (orthonormal)

$|v_i\rangle\langle v_i|$ → projector onto eigenvector v_i

Example:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$$

Eigenvalues

$$\boxed{\lambda_1 = 2, \quad \lambda_2 = 5}$$

Eigenvectors

$$|v_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |v_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|v_1\rangle\langle v_1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$|v_2\rangle\langle v_2| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$A = 2|v_1\rangle\langle v_1| + 5|v_2\rangle\langle v_2|$$

Substitute matrices:

$$= 2\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 5\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$$

Spectral Decomposition

Express operator using eigenvalues & projectors.

For Z:

$$Z = (+1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1|$$

Matrix form:

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

So:

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Expectation Value of an Operator

$$\boxed{\langle A \rangle = \langle \psi | A | \psi \rangle}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For the given state, find the expectation value of operator Z

Step: 1

$$Z|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Step: 2

$$\langle \psi | = \frac{1}{\sqrt{2}} (1 \;\; 1)$$

Step: 3

$$\langle Z \rangle = \frac{1}{2}(1 - 1) = 0$$

Average measurement of Z in state $|+\rangle$ is 0.

Expectation of Pauli-X in State $|0\rangle$

$$\boxed{\langle A \rangle = \langle \psi | A | \psi \rangle}$$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\langle 0| = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$X|0\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

$$\langle 0|X|0\rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0$$

Measuring X when the system is in $|0\rangle$ state:
Outcomes $+1$and -1occur with equal probability
Average = 0

Projection Operator

$$\boxed{P = |v\rangle\langle v|}$$

Example :

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$P_0 = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Now if we use projection operator on a given state then:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$P_0|\psi\rangle = \begin{pmatrix} \alpha \\ 0 \end{pmatrix}$$

Projective Measurement

Measurement in basis $\{|v_i\rangle\}$:

Projection operators:

$$P_i = |v_i\rangle\langle v_i|$$

Born Rule is given as: $\boxed{P(i) = \langle\psi|P_i|\psi\rangle = |\langle v_i|\psi\rangle|^2}$

After measurement:

$$|\psi\rangle \to \frac{P_i|\psi\rangle}{\sqrt{P(i)}}$$

This we got from postulate 3

Projective Measurement

Measurement in basis $\{|v_i\rangle\}$:

Projection operators:

$$P_i = |v_i\rangle\langle v_i|$$

$$\boxed{P(i) = \langle\psi|P_i|\psi\rangle = |\langle v_i|\psi\rangle|^2}$$

After measurement:

$$|\psi\rangle \to \frac{P_i|\psi\rangle}{\sqrt{P(i)}}$$

Example: State is given in ket.
Projectors are given. Find state in bra, multiply with projectors, and find the probability.

$$|\psi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$$

Projectors:

$$P_0 = \begin{pmatrix}1 & 0\\0 & 0\end{pmatrix}, \quad P_1 = \begin{pmatrix}0 & 0\\0 & 1\end{pmatrix}$$

Probabilities

$$P(0) = \frac{1}{2}, \quad P(1) = \frac{1}{2}$$

Measurement in X Basis/ Hadamard Basis

$$|+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$$

Projectors

$$P_+ = |+\rangle\langle+| = \frac{1}{2}\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}$$

$$P_- = \frac{1}{2}\begin{pmatrix}1 & -1\\-1 & 1\end{pmatrix}$$

Example: Measure $|0\rangle$ in X basis

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}$$

Compute:

$$P(+) = |\langle+|0\rangle|^2 = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

$$P(-) = \frac{1}{2}$$

Measurement in Y Basis

$$|+i\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \quad |-i\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}$$

Example: Measure $|0\rangle$ in Y basis

$$\langle +i|0\rangle = \frac{1}{\sqrt{2}}$$

$$P(+i) = \frac{1}{2}, \quad P(-i) = \frac{1}{2}$$

Measure Composite System

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix}$$

For outcome first qubit is 0

$$P_0 = |0\rangle\langle 0| \otimes I$$

$$|0\rangle\langle 0| = \begin{pmatrix}1 & 0\\0 & 0\end{pmatrix}, \quad I = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}$$

Tensor product:

$$P_0 = \begin{pmatrix}1 & 0\\0 & 0\end{pmatrix} \otimes \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} = \begin{pmatrix}1 & 0 & 0 & 0\\0 & 1 & 0 & 0\\0 & 0 & 0 & 0\\0 & 0 & 0 & 0\end{pmatrix}$$

Probability of outcome 0

$$P_0|\psi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\0\end{pmatrix} = \frac{1}{\sqrt{2}}|00\rangle$$

$$P(0) = \|P_0|\psi\rangle\|^2 = \frac{1}{2}$$

For outcome first qubit is 1

$$P_1 = |1\rangle\langle 1| \otimes I$$

$$= \begin{pmatrix}0 & 0\\0 & 1\end{pmatrix} \otimes I = \begin{pmatrix}0 & 0 & 0 & 0\\0 & 0 & 0 & 0\\0 & 0 & 1 & 0\\0 & 0 & 0 & 1\end{pmatrix}$$

Probability of outcome 1

$$P_1|\psi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\0\\0\\1\end{pmatrix} = \frac{1}{\sqrt{2}}|11\rangle$$

$$P(1) = \frac{1}{2}$$

For above step, remember the formula $\|v\|^2 = v^\dagger v$ Or compute this easily using $P(0) = \langle\psi|P_0^\dagger P_0|\psi\rangle$

Post Measurement step: Formula

$$\frac{P_0|\psi\rangle}{\sqrt{P(0)}}$$

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix}$$

For outcome: first qubit is 0

$$\boxed{|\psi_0\rangle = \frac{P_0|\psi\rangle}{\sqrt{P(0)}}}$$

$$= \frac{\frac{1}{\sqrt{2}}|00\rangle}{\sqrt{\frac{1}{2}}}$$

$$\sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$$

$$|\psi_0\rangle = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}|00\rangle = |00\rangle$$

For outcome: first qubit is 1

$$|\psi_1\rangle = \frac{\frac{1}{\sqrt{2}}|11\rangle}{\frac{1}{\sqrt{2}}} = |11\rangle$$

Example from Book :1

A system is in the state

$$|\psi\rangle = \frac{2}{\sqrt{19}}|u_1\rangle + \frac{2}{\sqrt{19}}|u_2\rangle + \frac{1}{\sqrt{19}}|u_3\rangle + \frac{2}{\sqrt{19}}|u_4\rangle + \sqrt{\frac{6}{19}}|u_5\rangle$$

where $\{|u_1\rangle, |u_2\rangle, |u_3\rangle, |u_4\rangle, |u_5\rangle\}$ are a complete and orthonormal set of vectors. Each $|u_i\rangle$ is an eigenstate of the system's Hamiltonian corresponding to the possible measurement result $H|u_n\rangle = n\varepsilon|u_n\rangle$, where $n = 1, 2, 3, 4, 5$.

(a) Describe the set of projection operators corresponding to the possible measurement results.

(b) Determine the probability of obtaining each measurement result. What is the state of the system after measurement if we measure the energy to be 3ε?

(c) What is the average energy of the system?

Answer

A system is in the state

$$|\psi\rangle = \frac{2}{\sqrt{19}}|u_1\rangle + \frac{2}{\sqrt{19}}|u_2\rangle + \frac{1}{\sqrt{19}}|u_3\rangle + \frac{2}{\sqrt{19}}|u_4\rangle + \sqrt{\frac{6}{19}}|u_5\rangle$$

$$p(a_i) = \left|\langle a_i|\psi_S(t)\rangle\right|^2 \quad \text{Born Rule,}$$

(a) The possible measurement results are ε, 2ε, 3ε, 4ε, and 5ε. These measurement results correspond to the basis states $|u_1\rangle$, $|u_2\rangle$, $|u_3\rangle$, $|u_4\rangle$, and $|u_5\rangle$, respectively. Hence the projection operators corresponding to each measurement result are

$$P_1 = |u_1\rangle\langle u_1|$$

$$P_2 = |u_2\rangle\langle u_2|$$

$$P_3 = |u_3\rangle\langle u_3|$$

$$P_4 = |u_4\rangle\langle u_4|$$

$$P_5 = |u_5\rangle\langle u_5|$$

Since the $|u_i\rangle$'s are a set of orthnormal basis vectors, the completeness relation is satisfied and

$$\sum_i P_i = I$$

Answer

A system is in the state

$$|\psi\rangle = \frac{2}{\sqrt{19}}|u_1\rangle + \frac{2}{\sqrt{19}}|u_2\rangle + \frac{1}{\sqrt{19}}|u_3\rangle + \frac{2}{\sqrt{19}}|u_4\rangle + \sqrt{\frac{6}{19}}|u_5\rangle$$

First we need to check and see if the state is normalized. This is done by calculating $\sum_{i=1}^{5}|c_i|^2$ and seeing if the result is 1. We have

$$\begin{aligned}\sum_{i=1}^{5}|c_i|^2 &= \left|\frac{2}{\sqrt{19}}\right|^2 + \left|\frac{2}{\sqrt{19}}\right|^2 + \left|\frac{1}{\sqrt{19}}\right|^2 + \left|\frac{2}{\sqrt{19}}\right|^2 + \left|\sqrt{\frac{6}{19}}\right|^2 \\ &= \frac{4}{19} + \frac{4}{19} + \frac{1}{19} + \frac{4}{19} + \frac{6}{19} \\ &= \frac{19}{19} = 1\end{aligned}$$

So, in the first case, applying the Born rule we have

$$\begin{aligned}\Pr(\varepsilon) = |\langle u_1|\psi\rangle|^2 &= \left|\langle u_1|\left(\frac{2}{\sqrt{19}}|u_1\rangle + \frac{2}{\sqrt{19}}|u_2\rangle + \frac{1}{\sqrt{19}}|u_3\rangle + \frac{2}{\sqrt{19}}|u_4\rangle + \sqrt{\frac{6}{19}}|u_5\rangle\right)\right|^2 \\ &= \left|\frac{2}{\sqrt{19}}\langle u_1|u_1\rangle + \frac{2}{\sqrt{19}}\langle u_1|u_2\rangle + \frac{1}{\sqrt{19}}\langle u_1|u_3\rangle + \frac{2}{\sqrt{19}}\langle u_1|u_4\rangle + \sqrt{\frac{6}{19}}\langle u_1|u_5\rangle\right|^2 \\ &= \left|\frac{2}{\sqrt{19}}(1) + \frac{2}{\sqrt{19}}(0) + \frac{1}{\sqrt{19}}(0) + \frac{2}{\sqrt{19}}(0) + \sqrt{\frac{6}{19}}(0)\right|^2 \\ &= \left|\frac{2}{\sqrt{19}}\right|^2 \\ &= \frac{4}{19}\end{aligned}$$

$$\Pr(i) = |\langle u_i|\psi\rangle|^2$$

The probability of obtaining the second measurement result is

$$\begin{aligned}\Pr(2\,\varepsilon) = |\langle u_2|\psi\rangle|^2 &= \left|\langle u_2|\left(\frac{2}{\sqrt{19}}|u_1\rangle + \frac{2}{\sqrt{19}}|u_2\rangle + \frac{1}{\sqrt{19}}|u_3\rangle + \frac{2}{\sqrt{19}}|u_4\rangle + \sqrt{\frac{6}{19}}|u_5\rangle\right)\right|^2 \\ &= \left|\frac{2}{\sqrt{19}}\langle u_2|u_2\rangle\right|^2 \\ &= \left|\frac{2}{\sqrt{19}}\right|^2 \\ &= \frac{4}{19}\end{aligned}$$

Answer

A system is in the state

$$|\psi\rangle = \frac{2}{\sqrt{19}}|u_1\rangle + \frac{2}{\sqrt{19}}|u_2\rangle + \frac{1}{\sqrt{19}}|u_3\rangle + \frac{2}{\sqrt{19}}|u_4\rangle + \sqrt{\frac{6}{19}}|u_5\rangle$$

$$P_3|\psi\rangle = (|u_3\rangle\langle u_3|)\left(|\psi\rangle = \frac{2}{\sqrt{19}}|u_1\rangle + \frac{2}{\sqrt{19}}|u_2\rangle + \frac{1}{\sqrt{19}}|u_3\rangle + \frac{2}{\sqrt{19}}|u_4\rangle + \sqrt{\frac{6}{19}}|u_5\rangle\right)$$

$$= |u_3\rangle\left(\frac{1}{\sqrt{19}}\langle u_3|u_3\rangle\right) = \frac{1}{\sqrt{19}}|u_3\rangle$$

Therefore

$$\Pr(3\varepsilon) = \langle\psi|P_3|\psi\rangle$$

$$= \left(\frac{2}{\sqrt{19}}\langle u_1| + \frac{2}{\sqrt{19}}\langle u_2| + \frac{1}{\sqrt{19}}\langle u_3| + \frac{2}{\sqrt{19}}\langle u_4| + \sqrt{\frac{6}{19}}\langle u_5|\right)\left(\frac{1}{\sqrt{19}}|u_3\rangle\right)$$

$$= \left(\frac{2}{\sqrt{19}}\right)\left(\frac{1}{\sqrt{19}}\right)\langle u_1|u_3\rangle + \left(\frac{2}{\sqrt{19}}\right)\left(\frac{1}{\sqrt{19}}\right)\langle u_2|u_3\rangle + \frac{1}{19}\langle u_3|u_3\rangle$$

$$+ \left(\frac{2}{\sqrt{19}}\right)\left(\frac{1}{\sqrt{19}}\right)\langle u_4|u_3\rangle + \left(\sqrt{\frac{6}{19}}\right)\left(\frac{1}{\sqrt{19}}\right)\langle u_5|u_3\rangle$$

$$= \frac{1}{19}$$

Similarly we find that

$$P_4|\psi\rangle = (|u_4\rangle\langle u_4|)|\psi\rangle = \frac{2}{\sqrt{19}}|u_4\rangle$$

$$P_5|\psi\rangle = (|u_5\rangle\langle u_5|)|\psi\rangle = \sqrt{\frac{6}{19}}|u_4\rangle$$

So we write

$$\Pr(4\varepsilon) = \langle\psi|P_4|\psi\rangle = \frac{4}{19}$$

$$\Pr(5\varepsilon) = \langle\psi|P_5|\psi\rangle = \frac{6}{19}$$

Answer

A system is in the state

$$|\psi\rangle = \frac{2}{\sqrt{19}}|u_1\rangle + \frac{2}{\sqrt{19}}|u_2\rangle + \frac{1}{\sqrt{19}}|u_3\rangle + \frac{2}{\sqrt{19}}|u_4\rangle + \sqrt{\frac{6}{19}}|u_5\rangle$$

(c) The average energy of the system is found using

$$\langle H\rangle = \sum_{i=1}^{5} E_i \langle\psi|P_i|\psi\rangle = \varepsilon\langle\psi|P_1|\psi\rangle + 2\varepsilon\langle\psi|P_2|\psi\rangle + 3\varepsilon\langle\psi|P_3|\psi\rangle$$

$$+ 4\varepsilon\langle\psi|P_4|\psi\rangle + 5\varepsilon\langle\psi|P_5|\psi\rangle$$

$$= \varepsilon\frac{4}{19} + 2\varepsilon\frac{4}{19} + 3\varepsilon\frac{1}{19} + 4\varepsilon\frac{4}{19} + 5\varepsilon\frac{6}{19}$$

$$= \frac{61}{19}\varepsilon$$

Example 2

A system is in the state

$$|\psi\rangle = \frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

(a) What is the probability that measurement finds the system in the state $|\phi\rangle = |01\rangle$?

(b) What is the probability that measurement finds the first qubit in the state $|0\rangle$? What is the state of the system after measurement?

Example

A system is in the state

$$|\psi\rangle = \frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

(a) Given that the system is in the state $|\psi\rangle$, the probability of finding it in the state $|\phi\rangle = |01\rangle$ is calculated using the Born rule, which is $\Pr = |\langle\phi|\psi\rangle|^2$. Since $\langle 0|1\rangle = \langle 1|0\rangle = 0$, we have

$$\begin{aligned}
\langle\phi|\psi\rangle &= \langle 01|\left(\frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle\right) \\
&= \frac{1}{\sqrt{8}}\langle 0|0\rangle\langle 1|0\rangle + \sqrt{\frac{3}{8}}\langle 0|0\rangle\langle 1|1\rangle + \frac{1}{2}\langle 0|1\rangle\langle 1|0\rangle + \frac{1}{2}\langle 0|1\rangle\langle 1|1\rangle \\
&= \sqrt{\frac{3}{8}}
\end{aligned}$$

Therefore the probability is

$$\Pr = |\langle\phi|\psi\rangle|^2 = \frac{3}{8}$$

Example

A system is in the state

$$|\psi\rangle = \frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

(b) To find the probability that measurement finds the first qubit in the state $|0\rangle$, we can apply $P_0 \otimes I = |0\rangle\langle 0| \otimes I$ to the state. So the projection operator P_0 is applied to the first qubit and the identity operator to the second qubit, leaving the second qubit unchanged. This obtains

$$\begin{aligned} P_0 \otimes I|\psi\rangle &= (|0\rangle\langle 0| \otimes I)\left(\frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle\right) \\ &= \frac{1}{\sqrt{8}}|0\rangle\langle 0|0\rangle \otimes |0\rangle + \sqrt{\frac{3}{8}}|0\rangle\langle 0|0\rangle \otimes |1\rangle + \frac{1}{2}|0\rangle\langle 0|1\rangle \otimes |0\rangle + \frac{1}{2}|0\rangle\langle 0|1\rangle \otimes |1\rangle \\ &= \frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle \end{aligned}$$

The probability of obtaining this result is

$$\begin{aligned} \Pr &= \langle\psi|P_0 \otimes I|\psi\rangle \\ &= \left(\frac{1}{\sqrt{8}}\langle 00| + \sqrt{\frac{3}{8}}\langle 01| + \frac{1}{2}\langle 10| + \frac{1}{2}\langle 11|\right)\left(\frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle\right) \\ &= \frac{1}{8} + \frac{3}{8} = \frac{1}{2} \end{aligned}$$

The state of the system after measurement

$$|\psi'\rangle = \frac{P_i|\psi\rangle}{\sqrt{\langle\psi|P_i|\psi\rangle}}$$

$$\begin{aligned} |\psi'\rangle &= \frac{\frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle}{\sqrt{\langle\psi|P_0 \otimes I|\psi\rangle}} = \sqrt{2}\left(\frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle\right) \\ &= \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|01\rangle \end{aligned}$$

Example 3

A three-qubit system is in the state

$$|\psi\rangle = \left(\frac{\sqrt{2}+i}{\sqrt{20}}\right)|000\rangle + \frac{1}{\sqrt{2}}|001\rangle + \frac{1}{\sqrt{10}}|011\rangle + \frac{i}{2}|111\rangle$$

(a) Is the state normalized? What is the probability that the system is found in the state $|000\rangle$ if all 3 qubits are measured?

(b) What is the probability that a measurement on the first qubit only gives 0? What is the postmeasurement state of the system?

$$|\psi\rangle = \left(\frac{\sqrt{2}+i}{\sqrt{20}}\right)|000\rangle + \frac{1}{\sqrt{2}}|001\rangle + \frac{1}{\sqrt{10}}|011\rangle + \frac{i}{2}|111\rangle$$

(a) To determine if the state is normalized, we compute the sum of the squares of the coefficients:

$$\begin{aligned}\sum_i |c_i|^2 &= \left(\frac{\sqrt{2}+i}{\sqrt{20}}\right)\left(\frac{\sqrt{2}-i}{\sqrt{20}}\right) + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{10}}\right)\left(\frac{1}{\sqrt{10}}\right) \\ &\quad + \left(\frac{i}{2}\right)\left(-\frac{i}{2}\right) \\ &= \frac{3}{20} + \frac{1}{2} + \frac{1}{10} + \frac{1}{4} = \frac{20}{20} = 1\end{aligned}$$

So the state is normalized. The probability the system is found in state $|000\rangle$ if all three qubits are measured is

$$\Pr(000) = \left(\frac{\sqrt{2}+i}{\sqrt{20}}\right)\left(\frac{\sqrt{2}-i}{\sqrt{20}}\right) = \frac{3}{20} = 0.15$$

$$|\psi\rangle = \left(\frac{\sqrt{2}+i}{\sqrt{20}}\right)|000\rangle + \frac{1}{\sqrt{2}}|001\rangle + \frac{1}{\sqrt{10}}|011\rangle + \frac{i}{2}|111\rangle$$

(b) The probability that a measurement on the first qubit is zero can be found by acting on the state with the operator $P_0 \otimes I \otimes I$ and computing $\langle\psi|P_0 \otimes I \otimes I|\psi\rangle$. This will project onto the $|0\rangle$ state for the first qubit while leaving the second and third qubits alone. We find that

$$\begin{aligned} P_0 \otimes I \otimes I|\psi\rangle &= \left(\frac{\sqrt{2}+i}{\sqrt{20}}\right)(|0\rangle\langle 0| \otimes I \otimes I)|000\rangle + \frac{1}{\sqrt{2}}(|0\rangle\langle 0| \otimes I \otimes I)|001\rangle \\ &\quad + \frac{1}{\sqrt{10}}(|0\rangle\langle 0| \otimes I \otimes I)|011\rangle + \frac{i}{2}(|0\rangle\langle 0| \otimes I \otimes I)|111\rangle \\ &= \left(\frac{\sqrt{2}+i}{\sqrt{20}}\right)|000\rangle + \frac{1}{\sqrt{2}}|001\rangle + \frac{1}{\sqrt{10}}|011\rangle \end{aligned}$$

The last term vanishes, since $\langle 0|1\rangle = 0$ So

$$\frac{i}{2}(|0\rangle\langle 0| \otimes I \otimes I)|111\rangle = \frac{i}{2}(|0\rangle\langle 0|1\rangle) \otimes |1\rangle \otimes |1\rangle = 0$$

Hence the probability that measurement on the first qubit finds 0 is

$$\langle\psi|P_0 \otimes I \otimes I|\psi\rangle = \left|\frac{\sqrt{2}+i}{\sqrt{20}}\right|^2 + \left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{1}{\sqrt{10}}\right|^2 = \frac{3}{20} + \frac{1}{2} + \frac{1}{10} = \frac{3}{4}$$

The postmeasurement state is

$$\begin{aligned} |\psi'\rangle &= \frac{P_0 \otimes I \otimes I|\psi\rangle}{\sqrt{\langle\psi|P_0 \otimes I \otimes I|\psi\rangle}} = \sqrt{\frac{4}{3}}\left(\left(\frac{\sqrt{2}+i}{\sqrt{20}}\right)|000\rangle + \frac{1}{\sqrt{2}}|001\rangle + \frac{1}{\sqrt{10}}|011\rangle\right) \\ &= \left(\frac{\sqrt{2}+i}{\sqrt{15}}\right)|000\rangle + \sqrt{\frac{2}{3}}|001\rangle + \sqrt{\frac{2}{15}}|011\rangle \end{aligned}$$

Example 4

A quantum system has a density matrix given by

$$\rho = \frac{5}{6}|0\rangle\langle 0| + \frac{1}{6}|1\rangle\langle 1|$$

What is the probability that the system is in the state $|0\rangle$?

If the measurement in question is described by a set of orthogonal projection operators $P_i = |u_i\rangle\langle u_i|$ corresponding to measurement result i then the probability of finding that measurement result is

$$\boxed{\Pr(i) = Tr(P_i^\dagger P_i \rho) = Tr(|u_i\rangle\langle u_i|\rho) = \langle u_i|\rho|u_i\rangle}$$

$$\begin{aligned}
\Pr(0) &= Tr(|0\rangle\langle 0|\rho) = \langle 0|\rho|0\rangle \\
&= \langle 0| \left(\frac{5}{6}|0\rangle\langle 0| + \frac{1}{6}|1\rangle\langle 1| \right) |0\rangle \\
&= \frac{5}{6}\langle 0|0\rangle\langle 0|0\rangle + \frac{1}{6}\langle 0|1\rangle\langle 1|0\rangle = \frac{5}{6}
\end{aligned}$$

Example 5

A system has the density operator

$$\rho = \frac{1}{3}|u_1\rangle\langle u_1| - i\frac{\sqrt{2}}{3}|u_1\rangle\langle u_2| + i\frac{\sqrt{2}}{3}|u_2\rangle\langle u_1| + \frac{2}{3}|u_2\rangle\langle u_2|$$

where the $|u_k\rangle$ constitute an orthonormal basis. What is the probability that a measurement finds the system in the state $|u_2\rangle$?

Solution

The projection operator corresponding to this measurement result is

$$P_2 = |u_2\rangle\langle u_2|$$

The probability is

$$\Pr(|u_2\rangle) = Tr(|u_2\rangle\langle u_2|\rho) = \langle u_2|\rho|u_2\rangle = \frac{2}{3}$$